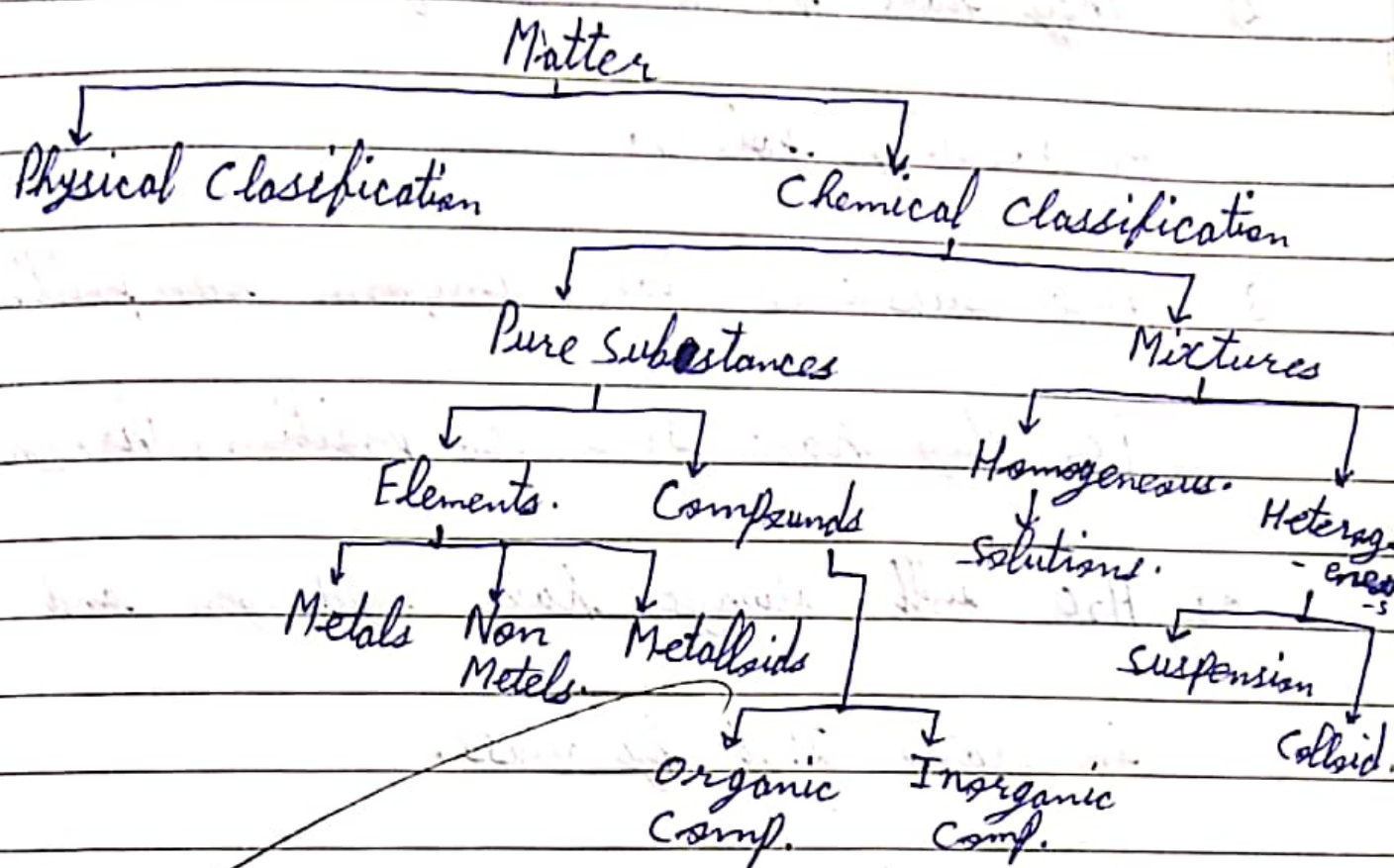


Is matter Around us pure:

Evergreen

Page No.

Date: / / 20



⇒ Pure Substances:

A pure substance is the substance that is made up of one type of particles (Atoms or Molecules).

- Some distinctive features of pure substances:

- 1) Pure substances are homogeneous irrespective of its origin or method of preparation.

2) They have fixed or definite boiling points or melting points.

3) Pure substance have uniform composition i.e; they have same composition throughout.

ex. H_2O will always have Hydrogen and Oxygen in ratio 2:16 by mass.

⇒ Element

- Term first used by Antoine Lavoisier.

⇒ Element is defined as a pure substance that is made up of only one type of atoms.

ex. Iron made up of iron atoms (Fe).

Copper made of copper (Cu) atoms.

* $\Rightarrow$ Total elements = 118

, Naturally occurring = 92

, Man made = 26.

$\Rightarrow$ Elements exist in all three forms i.e; solid, liquid and Gas.

$\rightarrow$ Liquid state - 2 elements i.e; Mercury (Hg)
Bromine (Br).

$\rightarrow$ Gas state $\rightarrow$ 11 elements - i.e; Oxygen (O_2)
Hydrogen (H_2)
Nitrogen (N_2)
Chlorine (Cl)
Fluorine (F).

$\rightarrow$ Noble gases :- Helium (He)
Neon (Ne)
Argon (Ar)
Krypton (Kr)
Xenon (Xe)
Radon (Rn).

→ Rest elements - Solids

⇒ Metals:

• 70 metals are naturally occurring and Aluminium

(Al) is the most abundant metal in earth's crust.

⇒ General Properties of metals:

1. Metals are solid in state except Mercury (Hg) that is liquid at room temperature.

2. Metals are malleable i.e; they can be drawn into thin sheets.

e.g Silver (Ag), Gold (Au), Aluminium (Al).

3. Metals are ductile i.e; they can be drawn into long wires.

Page No. _____
Date ____/____/____

e.g. Aluminium (Al), Copper (Cu).

4. Metals are good conductors of heat and electricity.

e.g. Aluminium (Al), Iron (Fe). Except lead (Pb) poorest conductor of heat.

5. Metals are lustrous i.e; have shiny appearance.

e.g. Gold (Au), Copper (Cu).

6. Metals are sonorous i.e; produce sound when hit by hard object. e.g. Aluminium.

7. Metals have high melting and boiling points.

8. Metals are generally hard in nature except sodium (Na), Potassium (K) that are soft and can be cut easily with a knife.

⇒ General Properties of Non-metals:-

- Only 18 elements are naturally occurring non-metals.

1) Non-metals exist as solid, Liquid and Gas.

e.g. Solid → Carbon, ~~sulphur~~ Sulphur.

Liquid → Bromine

Gas → Hydrogen, Oxygen, Chlorine etc.

2) Non-metals are neither malleable nor ductile.

3) They are non Lustrous i.e.; they do not have shiny appearance except diamond (Carbon).

4) Non metals are bad conductors of heat and electricity. Except Graphite which is good conductor of electricity.

⇒ Graphite is form of Carbon.

5) They are non sonorous i.e.; do not produce

sound when hit by hard object.

b) Non-metals have low melting and boiling points as compared to metals.

⇒ Metalloids

→ These are borderline elements. Their properties are similar to metals and some similar to non-metals.

e.g. Germanium, Silicon, Arsenic, Antimony

⇒ Compounds:-

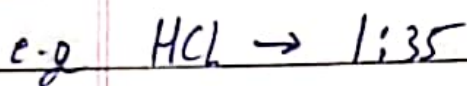
→ A compound is a pure substance that is formed by combination of two or more elements in fixed proportion by mass.

e.g. Water, Carbon dioxide, etc.



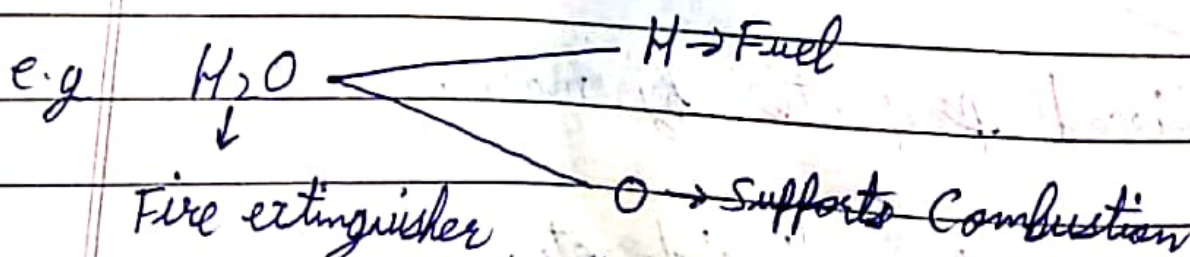
⇒ Properties of Compounds:

1) Elements combine in fixed proportion by mass to form compounds.



2) The properties of elements that combine together are entirely different from the compound formed.

'OR' - The properties of elements are lost when compound is formed.



- 3) The constituents of a compound can't be separated by simple physical processes like filtration, distillation, etc.
- 4) The compound has fixed melting and boiling point.
- 5) Compounds are homogeneous in nature i.e. composition is uniform.

- Types:

- Organic Compounds.
- Inorganic Compounds.

⇒ Organic Compounds:- Compounds containing Carbon along with other elements like H_2O , N , H , O , N , P , S .

→ They are obtained from living sources.

e.g. Carbohydrates, fats, Proteins, etc.

⇒ Inorganic Compounds:

→ Compounds obtained from non-living sources like minerals, etc.

e.g. Gypsum, Marble, Sodium Chloride, etc.

Nag 17/06/2016

⇒ Mixtures:

• Mixtures is a combination of two or more elements or compounds in any proportion.

e.g. Salt in Water, Oil and Water, etc.

→ Types of Mixture:

1) Homogeneous Mixtures

2) Heterogeneous Mixtures.

1) Homogeneous Mixture:-

- Types of mixture where components mix together completely and they can't be distinguished from each other.
- Composition is Uniform throughout.
- Only one phase present, there is no visible boundary of separation.

e.g. Salt in Water, Alloys.

Sugar in Water.

Ink in Water, etc.

2) Heterogeneous

- Mixture in which components do not mix completely and can be easily distinguished from each other.

→ Composition is not Uniform.

→ Visible boundary of separation in components and more than one phase is present.

e.g. Sand in Water
Oil in Water.
Petrol in Water, etc.

⇒ Properties of Mixtures:-

- 1) Mixtures can be homogeneous or heterogeneous.
- 2) Components of mixture are present in any ~~prop~~ proportion by mass.
- 3) They do not have a ~~fixed~~ fixed melting point or boiling point.
- 4) Properties of components are present in the final mixture.

5) No form of energy is absorbed or released when mixture is formed.

6) Components of mixtures can be easily separated by simple physical processes like filtration, distillation, etc.

⇒ Physical change and Chemical Change:-

- Physical Change:- During this change only physical properties like size, shape, colour, state etc, changes.

- No new substance is formed and composition of substance remains same.

→ It is mostly reversible.

Page No.
 Date / /
 e.g. Melting of ice, Cutting paper, dissolve sugar, in water etc.

⇒ Chemical Change: During this change composition of a substance changes. New substance with new properties formed.

⇒ Mostly irreversible.

e.g. burning of paper, Boiling of egg, Digestion of food, cooking of food, etc.

★ ⇒ Solutions:

• Solution is a homogeneous mixture of two or more chemically non-reacting substances where concentration can be varied upto certain

limits only.

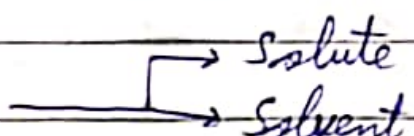
→ Solution can be Solid, Liquid or Gaseous.

eg Salt in water - Liquid solution.

Alloys - solid solution.

Air - Gaseous solution.

⇒ Components of Solution:

Two components 
→ Solute
→ Solvent.

DATE 20/06/2026

• Solute:-

• Component that is dissolved in any medium.

• It is present in lesser quantity.

• State of final solution and solute may or may not be same.

⇒ Solvent:- It is the medium in which solute is dissolved.

- It is present in larger quantity.
- The state of solvent and final solution is always same.

e.g. Sugar in Water, Sugar - Solute.
Water - Solvent.

⇒ Types of solution based on solvent:-

- 1) Aqueous Solution
- 2) Non-Aqueous Solution.

1) → Aqueous Solution. Solution in which solvent present is Water.

e.g. Salt in Water, alcohol in water, etc.

→ Non-Aqueous Solution. Solution having any solvent like ether, benzene, etc. other than water.

e.g. Carbon in Carbon disulphide (CS_2) (solvent).

⇒ Properties of Solutions

- 1) Solutions are homogeneous in nature.
- 2) Particles of solute easily pass through the filter paper.
- 3) Solutions are stable i.e; solute particles do not settle when kept undisturbed.
- 4) Properties of components i.e; solute and solvent retained in final solution.

⇒ Types of solution based on state of solvent and solute

	Solution	Solute	Solvent	Example.
1)	Solid Solution	Solid Liquid Gas	Solid Solid Solid	Alloys Hydrated salt. Gas trapped in minerals.
2)	Liquid Solution	Solid Liquid Gas	Liquid Liquid Liquid	
3)	Gaseous State	Solid Liquid Gas	Gas Gas Gas	

⇒ Concentration of the solution:-

- Amount of solute dissolved in given quantity of solvent or solution.

⇒ Based on Concentration → 3 Types of Solutions

~~Conc~~ Concentrated solution.

2) Dilute Solution

3) Saturated Solution.

1) Concentrated Solution

- Solution containing higher or large amount of solute.

2) Dilute Solution

- Solution containing very small amount of solute.

3) Saturated Solution

- Solution in which no more solute can be dissolved at room temperature.

→ Concentration of Solution can be expressed as:-

1) Mass Percentage or Mass by Mass percentage.

- Amount of solute in grams present in 100 grams.

of solution.

$$\Rightarrow \boxed{\text{Mass \%} = \frac{\text{Mass of Solute}}{\text{Mass of Solution}} \times 100}$$

2) Volume Percentage (Mass by volume Percentage)

- Amount of Solute in grams dissolved in 100 ml or 100 cm³ of solution.

$$\Rightarrow \boxed{\text{Volume \%} = \frac{\text{Mass of Solute}}{\text{Volume of solution}} \times 100}$$

3) Volume by Volume Percentage:-

- Volume of Solute present in 100 ml or 100 cm³ of solution.

$$\Rightarrow \boxed{\text{Volume by Volume \%} = \frac{\text{Vol. of Solute}}{\text{Vol. of Solution}} \times 100}$$

⇒ Suspension -

- It is a heterogeneous mixture in which particles of size greater than 100 nm (10^{-7} m) are distributed throughout a medium.
- Suspension consists of solid particles present in liquid medium.
- So suspension has two phases solid phase and liquid phase.
- e.g. Chalk powder in water, Turmeric in water, mud in water, paint, etc.

⇒ Properties

- 1) It is a heterogeneous mixture.
- 2) Size of particles in suspension is greater than

100 nm.

- 3) The particles of suspension do not pass through filter paper.
- 4) Suspension is Unstable i.e; particles settle at bottom when left undisturbed.
- 5) Particles of suspension can be seen with naked eye.
- 6) Suspension is opaque.

⇒ Colloids or Colloidal Solutions

Colloid is a heterogeneous mixture where particles of size range between 1 nm to 100 nm is distributed throughout the medium.

★ → It has two phases — $\left\{ \begin{array}{l} \rightarrow \text{Dispersed phase.} \\ \rightarrow \text{Dispersion.} \end{array} \right.$

eg Milk, blood, gum, Fog, Mist, shaving, cream, smoke.

→ Due to smaller particle size colloids appear as homogeneous but actually are heterogeneous.

1) Dispersed Phase:- The component ~~where~~ whose particles fall in range of 1nm to 100nm .

• It is also called as discontinuous ^{phase} ~~medium~~.

• This component is present in small proportion.

2) Dispersion Medium:- It is present in larger proportion and acts as medium in which dispersed phase is present.

→ Continuous phase.

eg Sulphur in Water.

↓
Dispersed
Phase

↓
Dispersion
Phase

⇒ Dispersed phase and dispersion medium can be solid, liquid and gas.

- Based on state of these phases types of colloidal solutions:

	Dispersed Phase	Dispersion Medium	Types	Example,
1.	Liquid	Gas	Aerosol	Fog, mist.
2.	Solid	Gas	Aerosol	Dust, smoke
3.	Gas	Liquid	Foam	Soap, Whipped Cream.
4.	Liquid	Liquid	Emulsion	Milk.
5.	Solid	Liquid	Sol	Gold sol
6.	Gas	Solid	Solid Foam	Pumice Sponge
7.	Liquid	Solid	Gel	Cheese, Butter
8.	Solid	Solid	Solid sol	Gem stones

⇒ Properties of Colloidal Solutions:-

1. They are heterogeneous in nature and consist of dispersed phase and dispersion medium.
2. Particle size falls in range of ~~1mm~~ 1nm - 100nm.
3. They easily pass through the filter paper.
4. They are stable and do not separate out when left undisturbed.
5. Brownian motion By Robert Brown.

When colloidal particles are seen under ultramicroscope they are seen to move in zig-zag motion which is called as Brownian motion or Pedesis.

6. Tyndal Effect. When a strong beam of light

is passed through colloidal solution in dark room, the path gets illuminated and this is called as Tyndal effect.

- It is because of scattering of light by colloidal particles.

~~Ng~~
1-8-2026